

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 16

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • A community maintained at a stage before the climax by regular human activity (grazing, mowing, burning, coppicing) that deflects succession (1) • Chalk grassland was produced by centuries of sheep grazing; without it succession to scrub and woodland resumes (1)	2
01.2	<i>1 mark for each point.</i> • Grazing removes taller / dominant grasses and prevents scrub establishing (5% cover) so small plants are not shaded out (1) • Trampling creates bare ground for wildflower seed germination and warm patches for invertebrates / reptiles (1) • Dung creates small nutrient-rich patches supporting different plants and invertebrates; sheep eat short grasses and some bushes (1) • Mowing removes everything at once including plants grazers avoid, and removes flowers / seeds before setting; burning removes vegetation but also kills invertebrates and allows scrub regrowth; abandoned plot dominated by scrub (55%) so few species (1)	4
01.3	<i>1 mark for each point.</i> • Sheep eat short grass (some breeds eat scrub); ponies avoid flowers but eat bracken; cattle remove long rough grass (1) • Choice depends on the vegetation to be controlled and the plants to be protected (1)	2
01.4	<i>1 mark for each point.</i> • Removes mature heather bushes and young trees that grazers will not eat, creating open unshaded ground (1) • Heat stimulates dormant heather seeds to germinate so young heather re-establishes (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3.	

Question	Marking guidance	Marks
02.1	<i>Correct answer only.</i> • B (1)	1
02.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Harvesting no faster than the MSY - resource never depleted (1) • Mixed species plantations - more food choices, relationships and niches (1) • Indigenous species - support indigenous insects, birds (1) • Mixed age structure - varied light, temperature, wind so more species; continuous cover and yield (1) • Selective logging - small clearings, less disturbance, long-term timber benefit (1) • Retention of dead wood, buffer zones, corridors (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369; AQA spec 3.5.3.	
03.1	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Hedgerows (1) • Hay meadows (1) • Grazed moorland / heathland (1) • Chalk grassland (1)	2
03.2	<i>1 mark for each point.</i> • Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive (1) • Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers (1)	2
03.3	<i>1 mark.</i> • The introduced (non-indigenous) predator may eat other species as well as the pest (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	
04.1	<i>1 mark for the correct answer.</i> • $(240\,000 - 2\,000) / 240\,000 \times 100 = 99.2\%$ (1)	1
04.2	<i>1 mark for each point.</i> • Whales are k-selected: slow growing, late maturing, one calf every few years, so populations recover very slowly (1) • Technology (harpoon cannons, steam / factory ships) allowed catches far above the sustainable yield within decades (1) • No regulation / open access to the ocean; whales were valuable (oil, spermaceti) and easily found (1)	3
04.3	<i>1 mark for each point.</i> • Moratorium on commercial whaling since 1986 until populations recover (1) • Total protection for certain species; whale sanctuaries (1) • Limits on numbers and sizes taken; protection of suckling mothers and calves (1) • Research into whale biology and threats (ship strikes, net entanglement); exceptions: aboriginal subsistence quotas (eg Inuit bowheads), 'scientific' permits (Japan - ruled not research in 2014), Norway / Iceland set their own quotas (1)	4

Question	Marking guidance	Marks
04.4	<i>1 mark for each point.</i> • ITTO's 1990 target of trade only from sustainably managed forests by 2000 was not met and little progress since (1) • Timber comes from many countries / private land with weak enforcement, high demand and alternative land uses; whaling involves few nations and a single treaty body (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 31-32; AQA spec 3.1.2.3.2.	
05.1	<i>1 mark for each point.</i> • Upwelling currents bring nutrients from the seabed to surface waters (1) • 24-hour daylight in summer allows a huge phytoplankton bloom (1) • Algae also grow under sea ice where krill shelter and feed; krill support most of the food web (1)	3
05.2	<i>1 mark for each point.</i> • Krill is the main food of baleen whales, penguins, seals and fish so over-harvesting removes food from most of the web (1) • Toothfish overfishing removes the food of sperm whales and elephant seals (1) • Long-lining for toothfish catches and drowns albatrosses on the hooks (by-catch) (1)	3
05.3	<i>1 mark for each point.</i> • Ice on land stores 70% of the Earth's fresh water, keeping sea levels low (1) • High albedo of ice reflects sunlight, limiting warming (reduction in ice area causes further warming) (1) • Carbon sequestration: carbon in algae eaten by krill sinks to the seabed in krill faeces (1)	3
05.4	<i>1 mark.</i> • Stronger winds spreading ice; more snowfall from warmer, moister air freezing onto slush; fresh meltwater from land forming a surface layer that freezes more easily (long-term trend still expected to be a decline) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 67-71; AQA spec 3.1.2.3.4.	
06.1	<i>1 mark.</i> • Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs (1)	1
06.2	<i>1 mark for each point.</i> • Negative feedback resists change / returns the system towards its previous state (1) • eg increased temperature increases cloud cover and albedo, or higher CO₂ increases photosynthesis (1) • Positive feedback increases the change / makes it self-sustaining (1) • eg melting permafrost releasing methane, declining albedo as ice melts, forest and peat fires (1)	4
06.3	<i>1 mark.</i> • Many species with a complex food web so the loss of one has a small effect; agroecosystems have few species and lack detritivores, pollinators and pest predators that maintain stability (1)	1

Question	Marking guidance	Marks
06.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Principles of natural systems: dynamic equilibria and negative feedback; diverse systems resistant to change; low energy-density renewable energy (solar-driven hydrological, carbon and nitrogen cycles); low-temperature processes using enzymes; material cycles with wastes as raw materials; small number of abundant elements; biodegradable non-toxic wastes - Human systems: non-renewable energy, high temperatures (Haber process, incineration, manufacturing), linear resource use, dispersal of minerals and mixtures that are hard to separate, persistent toxins (PCBs, organochlorines, toxic metals in electronics) - Applications: renewable energy and heat pumps, bioleaching, recycling and design for re-use, industrial symbiosis, organic and integrated agriculture, retaining hedgerows and pollinators, circular economy - Human responses that override feedback (fertilisers, irrigation, pesticides) and positive feedback triggered by human activity (permafrost, methane hydrates, fires); tipping points - Barriers: population size and affluence, attitudes of urban populations to distant resources, economics, need for international cooperation, existing infrastructure - Justified conclusion on how far natural principles can be applied	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-374; AQA spec 3.6.1-3.6.3.	
07.1	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Hydrological cycle (1) - Carbon cycle (1) - Nitrogen cycle (1) - Atmospheric circulation (1) - Thermohaline circulation (1) - Photosynthesis (1)	2
07.2	<i>1 mark for each point.</i> - Enzymes reduce the activation energy of reactions (1) - So reactions such as photosynthesis, decomposition and nitrogen fixation do not need high temperatures (1)	2
07.3	<i>Any 1 from the points below - 1 mark each, maximum 1.</i> - Haber process (nitrogen fixation) (1) - Incineration of wastes (decomposition) (1) - Many manufacturing processes (1)	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
08.1	<i>1 mark for each point.</i> - As sea otters declined (1960-1980) sea urchins increased (to 350) and kelp area fell (to 25) (1) - As otters recovered after 1990 urchins declined and kelp recovered (1) - Otter and kelp trends are positively correlated; urchins negatively correlated with both (1)	3
08.2	<i>1 mark for each point.</i> - Sea otters are predators of sea urchins so fewer otters allowed the urchin population to rise (1) - Sea urchins graze kelp so the kelp forest was over-grazed and destroyed (1) - Kelp forests are habitat / food for many species including otters (keystone species / trophic cascade); otter recovery restored the balance (1)	3
08.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Over-fishing of sandeels reduced puffins (1) - Over-collection of turtle eggs reduced turtles so jellyfish (their food) increased; over-fishing of sardines / herring (competitors) also increased jellyfish (1) - Loss of pollinating bees (pesticides, wild flower loss) reduces plant reproduction; loss of elephants / large herbivores reduces seed dispersal of trees (1)	2
08.4	<i>1 mark for each point.</i> - They control abiotic features: beaver dams create ponds colonised by aquatic species; elephants create clearings and keep water holes open (1) - Species that depend on these features (light, water, minerals) cannot survive in their absence (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 23-25; AQA spec 3.1.2.2.	
09.1	<i>1 mark for each point.</i> - Chicken and salmon need far less feed per kg gained (1.2-1.8 vs 7.0) (1) - Fish are ectotherms so no energy is used for thermoregulation and water supports their mass; poultry grow fast and are kept warm indoors with movement restricted (1) - Cattle are large endotherms, use energy in movement, and lose energy in methane from cellulose digestion (1)	3
09.2	<i>1 mark for each point.</i> - Control of movement (housing / restricted grazing) so less energy is used in muscle activity (1) - Temperature control (heated / insulated buildings) so less energy is used in maintaining body temperature (1) - Species selection: choose species with low FCRs (poultry, fish); high-energy easily digested diets; slaughter at the fastest growth stage (1)	3

Question	Marking guidance	Marks
09.3	<i>1 mark for each point.</i> • Only about 10% of the energy at one trophic level is passed to the next; most is lost as heat in respiration / metabolism or in undigested material (1) • Humans eating plants (primary consumers) receive more of the energy captured by photosynthesis than eating animals fed on those plants (secondary consumers) (1) • Livestock better where land cannot grow crops (upland, semi-arid, steep, thin soils): grazing animals with symbiotic gut bacteria digest cellulose in grass that humans cannot (1) • Omnivores (pigs) convert food wastes (dairy by-products, catering waste) into meat (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	
10.1	<i>1 mark for each point.</i> • Plants produce chemicals (often alkaloids) that are toxic to the herbivores that would eat them (1) • In carefully controlled amounts these chemicals have beneficial physiological effects in humans (1)	2
10.2	<i>1 mark for each point.</i> • Only a very small proportion of species have been studied for medicinal substances, so it is not known which species will be useful (1) • Conserving the habitat protects all the species in it, including those not yet studied and the species they depend on (1)	2
10.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Marsupials (kangaroo / wallaby): young develop in the pouch so are easier to study than a human foetus - developmental problems (1) • Hippopotamus: hipposudoric acid skin secretion (sunscreen / antimicrobial) - burns treatment (1) • Marine sponges: proteins preventing graft rejection - organ transplants (1)	2

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10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Resources: wood, fibres, oils, fuels, food; new food species adapted to local conditions (morama bean, yeheb, perennial maize, Kernza, bison, eland) • Biomimetics: whale flipper tubercles, shark skin, termite-mound ventilation, bird bones, gecko adhesion, burrs / Velcro, lotus self-cleaning, spider silk • Medicines and physiological research; genetic resources - crop wild relatives, Vavilov centres, gene pools, inbreeding • Ecosystem services: atmospheric composition, hydrological cycle, biogeochemical cycles, soil maintenance; interspecies relationships - food, pollination, seed dispersal, habitat provision • Evaluation: services go unnoticed because of dynamic equilibrium; unknown future value; economic vs moral / intrinsic arguments; conservation also needed for aesthetic and cultural reasons • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Question	Marking guidance	Marks
11.1	Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Species selection: market, range of tolerance, controllability • Abiotic control: temperature, light, water, soil fertility / nutrients, aeration, salinity, pH, CO₂, topography, relief, wind - methods and examples • Biotic control: pest groups, cultural control, biological control (successes and failures), sterile males, pheromones, genetic resistance, pesticides (properties), antibiotics, pollinators, soil biota • Manipulation of food species: population density, monocultures, selective breeding, asexual reproduction, cross-breeding, AI / embryo transfer, GM; energetics - intensive vs extensive, energy subsidies, ratios, FCR, food chain losses • Impacts: habitat loss / change, reduced biodiversity, introduced species, pesticide and nutrient pollution, GHGs, hydrological changes, soil erosion; positive: plagioclimax habitats • Social, economic and political influences shaping methods • Sustainable strategies and evaluation; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 306-337; AQA spec 3.5.1.	25

Question	Marking guidance	Marks
11.2	Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Setting priorities: IUCN Red List criteria, EDGE, endemic, keystone, flagship species, population dispersal • Legislation and protocols: Wildlife and Countryside Act, designated areas (SSSI to MCZ), CITES Appendices, IWC, EU CFP, ITTO - successes and limits of enforcement • Captive breeding and release: selection, difficulties, techniques, release site criteria, hard / soft release, monitoring; seed banks • Habitat conservation: creation, design (area, shape, corridors, age structure, colonisation), abiotic and biotic management, plagioclimax maintenance, population control, reintroductions • Habitat case studies (woodland, rainforest, coral reefs, islands, mangroves, Antarctica) illustrating threats and conservation efforts • Monitoring and new technologies informing decisions • Evaluation: examples of success (Iberian lynx, Hawaiian goose, Scilly rats, Yellowstone) and failure (ITTO, tigers); cost, politics, climate change; judgement on the most effective combination Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-71; AQA spec 3.1.2.3.	25